

ADR-01: Business Use Case Selection for Customer Query Analytics

Architecture Decision Record

September 8, 2025

1 Status

Accepted

2 Context

Customer support operations face critical challenges in ticket routing, annotator quality assessment, and understanding customer pain points. Multiple potential use cases were considered for implementing a comprehensive customer query analytics system:

2.1 Business Problem

- Support tickets frequently misrouted due to unclear classification criteria
- Annotator performance varies significantly, impacting downstream operations
- Customer sentiment and emerging issues remain invisible to management
- Manual routing creates bottlenecks and delays response times
- No systematic approach to identifying customer risk patterns

2.2 Alternative Use Cases Considered

1. **E-commerce Product Reviews Analysis:** Focus on product feedback and recommendations
2. **Social Media Sentiment Monitoring:** Track brand perception across social platforms
3. **Employee Feedback Analysis:** Internal HR sentiment and engagement tracking
4. **Customer Support Ticket Classification:** Hierarchical ticket routing and quality assessment
5. **Sales Call Transcription Analysis:** Lead qualification and sales process optimization

3 Decision

We selected **Customer Support Ticket Classification** as the primary use case for the following reasons:

3.1 Strategic Alignment

- Direct impact on customer experience and satisfaction metrics
- Clear ROI through reduced response times and improved routing accuracy
- Addresses immediate operational pain points affecting daily business operations
- Scalable solution applicable across multiple product lines and support channels

3.2 Technical Feasibility

- Well-defined hierarchical classification structure (L1: Technical/Billing/Account, L2: Specific subtypes)
- Measurable success criteria through inter-annotator agreement analysis
- Rich dataset enabling comprehensive analytics across multiple dimensions
- Clear stakeholder requirements from support management, QA teams, and operations

3.3 Business Impact

- **Operational Efficiency:** Automated routing reduces manual triage time by estimated 60%
- **Quality Assurance:** Systematic annotator performance assessment enables targeted training
- **Customer Insights:** Sentiment analysis and risk detection provide proactive issue identification
- **Strategic Intelligence:** Topic modeling reveals emerging customer needs and product gaps

4 Consequences

4.1 Positive Outcomes

- **Immediate Value:** Support team can implement improved routing within weeks
- **Measurable Results:** Clear KPIs (routing accuracy, response time, customer satisfaction)
- **Scalable Framework:** Architecture extends to other text classification challenges
- **Data-Driven Decisions:** Management gains visibility into customer patterns and annotator performance

4.2 Trade-offs and Limitations

- **Domain Specificity:** Solution optimized for support tickets, requiring adaptation for other use cases
- **Annotation Dependency:** Quality heavily dependent on consistent labeling policies and annotator training

- **Complexity Management:** Multi-dimensional analysis (sentiment, topics, entities) requires careful prioritization
- **Change Management:** Success requires buy-in from support staff and process modifications

4.3 Rejected Alternatives and Rationale

- **E-commerce Reviews:** Lower business priority, existing solutions available
- **Social Media Monitoring:** Requires different technical stack, less direct ROI
- **Employee Feedback:** Sensitive data concerns, different stakeholder requirements
- **Sales Call Analysis:** Technology readiness gaps in speech-to-text pipeline

5 Implementation Guidelines

5.1 Success Metrics

- Krippendorff's $\alpha \geq 0.67$ for acceptable annotation reliability
- Routing accuracy improvement of $\geq 25\%$ over baseline manual classification
- Customer satisfaction score improvement of $\geq 10\%$ through faster resolution
- Annotator consistency improvement measured through reduced pairwise disagreement

5.2 Key Stakeholders

- **Primary:** Customer Support Management, QA Teams
- **Secondary:** Product Teams, Operations Management
- **Governance:** Data Science Team, IT Architecture

5.3 Critical Dependencies

- Consistent labeling policy development and maintenance
- Annotator training program aligned with classification schema
- Integration capabilities with existing ticketing systems
- Change management support for workflow modifications

6 Review and Evolution

This decision should be reviewed quarterly based on:

- Business value delivery and KPI achievement
- Technical performance and scalability requirements
- Changing business priorities and customer needs

- Lessons learned from implementation and operation

Future evolution may extend the framework to adjacent use cases such as chat transcripts, email inquiries, or product feedback analysis.